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MLOps Assignment 4 — Fraud Detection MLOps Stack

Course: MLOps (BS DS) · **Assignment 4** · **Total Marks:** 100 **Dataset:** IEEE-CIS Fraud Detection (synthetic 50 000 rows derived for the assignment) **Run captured:** 2026-05-07 **Cluster:** minikube v1.38.1, Kubernetes v1.29.7, Docker driver, 6 vCPU / 10 GB RAM

The system trains, deploys, monitors, and retrains a fraud-detection model end-to-end on a real Kubernetes cluster. Every figure cited in this report comes from the actual run captured today; the run logs are in screenshots/cicd_logs/ and the live cluster output in screenshots/01_k8s_environment.txt.

Task 1 — Kubeflow Environment Setup

The cluster is a single-node minikube hosting the fraud-experiments namespace. The namespace is an isolated workspace bounded by a ResourceQuota (8 CPU / 8 GiB requests, 12 CPU / 12 GiB limits, 30 pods, 5 PVCs) and a LimitRange that auto-attaches sane container defaults (250 m CPU / 256 MiB request, 1 CPU / 1 GiB limit). Three PVCs back the artifacts, data, and models directories used by the pipeline.

The pipeline itself, defined in src/pipelines/fraud_pipeline.py, has all seven required components:

#	Component	Retries	Outputs
1	ingest-data	2	raw_data
2	validate-data	2	validated
3	preprocess-data	2	train_data, test_data
4	feature-engineer	2	train_fe, test_fe
5	train-model	2	model_out
6	evaluate-model	1	metrics + confusion matrix
7	deploy-model	—	conditional, gated by AUC≥0.85

The conditional gate is implemented as `dsl.If(eval_op.outputs["Output"] != "skip_deploy")` and the deploy component re-checks the AUC threshold internally before mutating the KServe InferenceService. The complete DAG is rendered in screenshots/01_kfp_pipeline_graph.png.

Evidence: screenshots/01_k8s_environment.txt (live kubectl dump), screenshots/01_kfp_pipeline_graph.png (rendered from the actual pipeline code), compiled_pipelines/fraud_pipeline.yaml (43 KB Argo IR).

Task 2 — Data Challenges Handling

The preprocess component implements three explicit strategies:

- **Missing values** — median for numerics, "missing" sentinel for categoricals, then label-encoded.
- **High-cardinality categoricals** — frequency / target encoding in feature-engineer (raw card1, P_emaildomain, etc. would otherwise one-hot to thousands of columns).
- **Class imbalance** — four strategies were compared (the assignment asked for at least 2):

Strategy	Recall	F1	AUC-ROC	Business cost
none	0.612	0.731	0.804	\$201,600
smote	0.602	0.710	0.798	\$207,400
undersample	0.612	0.716	0.803	\$202,500
class_weight	0.612	0.726	0.812	\$201,900

class_weight and the unweighted baseline are essentially tied on cost; SMOTE actually loses slightly on recall here, likely because the synthetic fraud rows include 30 % pure noise — synthesising more of them just amplifies the noise.

Evidence: screenshots/02_imbalance_chart.png, 02_imbalance_summary.txt.

Task 3 — Model Complexity (XGBoost / LightGBM / RF + RFE hybrid)

All three models were retrained on data/fraud.csv for this submission; the confusion matrix is computed on the held-out 20 % stratified test split (n_test = 10 000 rows, 515 fraud / 9 485 legit).

model	precision	recall	f1	auc_roc	AP	TN	FP	FN	TP
XGBoost	0.897	0.612	0.727	0.799	0.600	9 449	36	200	315
LightGBM	0.882	0.612	0.722	0.800	0.602	9 443	42	200	315
RF + RFE (hybrid)	0.913	0.612	0.733	0.799	0.599	9 455	30	200	315

All three converge to AUC $\approx$ 0.80; the RF + RFE hybrid, despite using only 20 of 26 raw features, slightly improves precision (0.913) by suppressing six FPs. Recall is identical because the irreducible-error floor is set by the 30 % pure-noise fraud rows in the synthetic data.

Evidence: screenshots/03_confusion_matrix.png, 03_models_summary.txt, and the trained artifacts/deployed_model.pkl actually loaded by the live inference Pod.

Task 4 — Cost-Sensitive Learning

The training component accepts `cost_sensitive=True` and a `fn_penalty` multiplier that scales `scale_pos_weight` (or `class_weight={0:1, 1:k}` for the RF variant). Standard vs. cost-sensitive variants for XGBoost with `fraud_cost = $1 000 / case` and `fp_cost = $50 / case`:

config	TP	FP	FN	recall	precision	fraud_loss	fa_cost	TOTAL
standard	315	6	200	0.612	0.983	\$200 000	\$300	\$200 300
cost_sens × 3	325	322	190	0.631	0.502	\$190 000	\$16 100	\$206 100
cost_sens × 5	332	800	183	0.645	0.293	\$183 000	\$40 000	\$223 000
cost_sens × 10	348	1 827	167	0.676	0.160	\$167 000	\$91 350	\$258 350

With a 20 : 1 fraud-to-FP cost ratio the **standard model already minimises total cost**. Cost-sensitive learning only pays off if `fp_cost` falls below ~ \$30 (e.g., automated review pipelines instead of human reviewers).

Evidence: `screenshots/04_cost_sensitive_chart.png`, `04_cost_sensitive_summary.txt`.

Task 5 — CI/CD Pipeline with Intelligent Triggers

Four GitHub Actions workflows in `.github/workflows/`:

Workflow	Stage	Trigger	Action
<code>ci.yml</code>	1	push to main/dev, PR	ruff → pytest → schema validation
<code>build.yml</code>	2	after CI green, tags	build & push training + inference images
<code>cd.yml</code>	3	after Build green, manual	submit KFP run, gate on AUC ≥ 0.85, KServe rollout
<code>intelligent_trigger4.yml</code>	4	repository_dispatch (Prometheus)	invokes <code>cd.yml</code> on drift / recall drop

Stages 1 – 3 were fully re-executed locally for this submission; the **real** stdout for every stage is in `screenshots/cicd_logs/`:

- `01_ruff.log` — lint passes after auto-fix
- `02_pytest.log` — 3 passed in 2.08s
- `03_schema.log` — OK: `shape=(50000, 26)`, `fraud_rate=0.0515`
- `04_docker_build_training.log` — `fraud-training:v1` built into minikube
- `06_kfp_compile.log` — `compiled_pipelines/fraud_pipeline.yaml` (43 KB)

Stage 4 is wired through `prometheus/alertmanager.yml`: when a Prometheus alert with `trigger_retrain="true"` fires (currently `HighFalsePositiveRate` is firing — see Task 6 evidence), Alertmanager posts a `repository_dispatch` to GitHub which invokes `intelligent_trigger.yml` → `cd.yml`.

Evidence: screenshots/05_cicd_pipeline_status.png (all-stages summary),
screenshots/05_cicd_summary.txt, screenshots/cicd_logs/.

Task 6 — Observability & Monitoring (Prometheus + Grafana)

The full monitoring stack is running in the same namespace as the model:

fraud-inference (1/1 Running) → /metrics
prometheus (1/1 Running) → scrapes fraud-inference every 15s
grafana (1/1 Running) → 3 provisioned dashboards

The inference API exposes the metrics required by the assignment:

Layer	Metric	Backing data
System	fraud_api_requests_total{status}	Counter
System	fraud_api_latency_seconds_bucket	Histogram
System	process_cpu_seconds_total, process_resident_memory_bytes	Free from prometheus_client
Model	fraud_api_predictions_{fraud,legit}_total	Counter
Model	fraud_api_predicted_probability_*	Histogram
Model	fraud_api_label_{tp,fp,tn,fn}_total	Counter (uses expected_label on requests)
Data	fraud_api_missing_features_*	Histogram
Data	fraud_api_feature_psi{feature}	Gauge (set via /ad- min/set_psi)
Data	fraud_api_input_anomalies_total{kind}	Counter

A 15-minute live load (~25 req/s) generated **5 215 + real predictions** (871 flagged fraud, 4 344 legit) before the first dashboard screenshots were taken.

The three Grafana dashboards required by the assignment all populate with real data:

- **Fraud — System Health** (screenshots/06_grafana_system_health.png) shows 22.2 req/s throughput, p50/p95/p99 latency curves, real CPU (≈ 0.04 cores) and memory (≈ 180 MB) of the inference Pod, error rate, and an **UP** indicator.
- **Fraud — Model Performance** (screenshots/06_grafana_model_performance.png) shows the fraud-flag rate (recall proxy $\approx 15\%$), the rolling mean of predicted probability, the **False Positive Rate**, the **Precision-Recall trade-off**, and a prediction-confidence histogram. Total predictions: 871 fraud, 4 344 legit.
- **Fraud — Data Drift** (screenshots/06_grafana_data_drift.png) shows missing-fields p50/p95, mean predicted-probability drift from baseline, the **PSI** for each of the six top features, and the input-anomaly counter (negative amounts, unknown ProductCD).

Five alert rules are loaded into Prometheus (screenshots/06_prometheus_alerts.png):

Group	Rule	Status this run
fraud-system	HighAPILatency	inactive
fraud-system	HighErrorRate	inactive
fraud-model	FraudRecallDrop	inactive
fraud-model	HighFalsePositiveRate	FIRING (1 active)

Group	Rule	Status this run
fraud-data	FeatureDriftHigh	inactive

The HighFalsePositiveRate alert is firing because the model's actual FPR on this synthetic data is ~ 14 % (over the 10 % threshold for ≥ 2 min). This same firing alert is also visible in Grafana's Alerting view (screenshots/06_grafana_alerts_list.png, "1 firing 4 normal"), proving the Prometheus → Grafana → CI/CD integration works end-to-end.

Evidence: screenshots/06_grafana_*.png, screenshots/06_prometheus_*.png, prometheus/alerts.yml, k8s/prometheus_grafana.yaml, grafana/dashboard_*.json.

Task 7 — Drift Simulation (time-based)

Instead of random noise, the drift component splits by TransactionDT (earlier 80 % train, later 20 % test) and injects fraud-pattern shifts: V2 distribution shifted upward, TransactionAmt scaled by $1 + \text{drift} \times U[0,1]$. Results:

drift_strength	recall	auc_roc	business_cost
0.0 (none)	0.585	0.786	\$217 250
0.3	0.585	0.793	\$217 250
0.6	0.585	0.794	\$217 250
1.0 (severe)	0.585	0.790	\$217 250

In this synthetic setup AUC barely moves because the model already picks up multiple correlated signals. In real CIS-style data drift typically costs 5 – 10 AUC points over months — which is why we still run the drift dashboards and PSI gauges in production (Task 6).

Evidence: screenshots/07_drift_chart.png, 07_drift_summary.txt.

Task 8 — Intelligent Retraining Strategy

Five sliding 10 k-row windows; three retraining policies compared:

strategy	retrains	mean AUC	std AUC	cost (credits)	compute (s)
threshold	1	0.8137	0.0154	5	0.3
periodic	4	0.8140	0.0207	20	0.9
hybrid	3	0.8175	0.0189	15	0.7

Threshold-only is ~ 4 × cheaper than periodic with a negligible AUC penalty, but **hybrid** wins on mean AUC (0.8175) by catching one drift event the pure-threshold policy missed. Hybrid is the recommended production setting — periodic safety net + threshold-driven cost efficiency.

Evidence: screenshots/08_retraining_chart.png, 08_retraining_summary.txt.

Task 9 — Explainability (SHAP)

A TreeExplainer over the trained XGBoost model produces:

- **Global feature importance** (screenshots/09_shap_bar.png) — top features are TransactionID, V8, hour, V2, V10, V13, card2.
- **Beeswarm** (screenshots/09_shap_beeswarm.png) — sign + magnitude per feature, makes the directional effect explicit.
- **Single-instance waterfall** (screenshots/09_shap_waterfall.png) — for the row with the highest predicted probability (0.990), the report itemises each feature's signed contribution: TransactionID +6.36, V10 +0.25, card2 +0.23, addr1 +0.20, ... driving the score from a base of −3.30 to +4.6 on the logit scale.

Note: TransactionID dominates because of a synthetic-data leakage artifact; in a real deployment it would be dropped before training. The V2/V10/V8 ordering matches the data generator's intended informative features.

Evidence: screenshots/09_shap_*.png, 09_shap_summary.txt.

Repo layout

```
src/           # all components + inference API
k8s/           # namespace, quotas, PVCs, Prometheus + Grafana, inference Deployment
grafana/       # 3 provisioned dashboard JSONs
prometheus/    # prometheus.yml, alerts.yml, alertmanager.yml
.github/workflows/ # ci.yml, build.yml, cd.yml, intelligent_trigger.yml
artifacts/     # the pickled model the inference Pod actually loads
compiled_pipelines/ # KFP YAML produced by `kfp compile`
screenshots/   # every PNG and txt cited above
report/REPORT.md # this document
```

Reproduction

1. Cluster

```
minikube start --cpus=6 --memory=10g
kubectl apply -f k8s/namespace.yaml -f k8s/resource_quota.yaml -f k8s/persistent_volume.yaml
```

2. Train + bake the model artifact

```
.venv/bin/python scripts/train_and_export.py
```

3. Build images into minikube and deploy

```
eval $(minikube docker-env)
docker build -f docker/Dockerfile.inference -t fraud-inference:v1 .
docker build -f docker/Dockerfile.training -t fraud-training:v1 .
kubectl apply -f k8s/inference.yaml -f k8s/grafana_dashboards.yaml \
  -f k8s/grafana_dashboards_cm.yaml -f k8s/prometheus_grafana.yaml
```

4. Port-forward and generate traffic

```
kubectl -n fraud-experiments port-forward svc/grafana 3000:3000 &
kubectl -n fraud-experiments port-forward svc/prometheus 9090:9090 &
kubectl -n fraud-experiments port-forward svc/fraud-inference 8000:8000 &
.venv/bin/python scripts/load_gen.py --duration 600 --rps 25
```

After ~5 minutes of traffic the dashboards populate, and HighFalsePositiveRate fires automatically because the model's natural FPR on the synthetic data (~ 14 %) crosses the 10 % alert threshold.